My Study Notes

Timóteo

June 28, 2026

Contents

1 Introduction

Welcome to my study notes. This document is organized into multiple chapters to keep everything clean and modular.

2 Conjectures and Theorems

2.1 Basic Definitions

Definition 1. *A* conjecture *is a mathematical statement that is believed to be true based on evidence or intuition, but which has not yet been proved or disproved.*

Definition 2. *A* theorem *is a mathematical statement that has been rigorously proved to be true from previously established results (such as axioms, definitions, and other theorems).*

2.2 Examples from Number Theory

Example 2.1 (A true conjecture that became a theorem)**.** *Historically, mathematicians* conjectured *that there are infinitely many prime numbers. This was later proved by Euclid and is now a theorem:*

Theorem 1 (Euclid)**.** *There are infinitely many prime numbers.*

Thus, a statement can start as a conjecture and, once proved, become a theorem.

Example 2.2 (A conjecture that is still open)**.** Goldbach's Conjecture *is one of the most famous unsolved problems in number theory:*

Conjecture 1 (Goldbach)**.** *Every even integer $n \geq 4$ can be written as the sum of two prime numbers.*

This statement has been checked for very large values of n, but no complete proof or counterexample is known.

Example 2.3 (A false conjecture)**.** *Consider the following statement:*

Conjecture 2. *Every odd integer $n > 1$ is a prime number.*

At first glance, small examples like $3, 5, 7$ might suggest this is true. However, it is false*. For instance,*

$$9 = 3 \cdot 3, \quad 15 = 3 \cdot 5,$$

so 9 and 15 are odd but not prime. These are counterexamples that disprove the conjecture.

2.3 Remarks

Remark 1. *Conjectures play a central role in mathematical research. They guide exploration and often lead to new techniques, whether they eventually become theorems or are disproved by counterexamples.*

3 Logical Implication, Necessary Conditions, and Sufficient Conditions

3.1 Introduction

Logic is the language we use to organize mathematical reasoning. Before proving theorems, we need to understand how mathematical statements are connected.

A *statement* is a sentence that is either true or false, but not both.

For example,

"2 is even"

is a true statement, while

"3 is even"

is a false statement.
However,

$$x + 1$$

is not a statement by itself, because it is not true or false. It is only an expression. On the other hand,

$$x + 1 > 0$$

can become a statement once we say what values of (x) we are considering.

In this chapter we introduce implication, necessary conditions, sufficient conditions, logical equivalence, truth tables, and the contrapositive.

3.2 Implication

Let (A) and (B) be statements.

Definition 3.

Implication

The statement

$$A \Rightarrow B$$

means:

if A is true, then B must be true.

We read this as:

"A implies B"

or

"if A, then B."

The implication ($A \Rightarrow B$) does not say that (A) causes (B). It only says that it is impossible for (A) to be true while (B) is false.

So the meaning of

$$A \Rightarrow B$$

is:

there is no case where A is true and B is false.

When (A) is false, the implication makes no demand on (B). For this reason, in classical logic the implication ($A \Rightarrow B$) is considered true whenever (A) is false.

3.3 Necessary and Sufficient Conditions

Implication allows us to describe when one condition guarantees another.

Definition 4.

$$SufficientCondition$$

A statement (A) is a sufficient condition *for (B) if*

$$A \Rightarrow B.$$

This means that (A) being true is enough to guarantee that (B) is true.

In other words, if we know (A), then we automatically know (B).

Definition 5.

$$NecessaryCondition$$

A statement (B) is a necessary condition *for (A) if*

$$A \Rightarrow B.$$

This means that whenever (A) is true, (B) must also be true. Without (B), (A) cannot happen.

Therefore, the same implication

$$A \Rightarrow B$$

has two interpretations:

- (A) is sufficient for (B);
- (B) is necessary for (A).

This is one of the most important ideas in logic.

3.4 Example: Divisibility

Let

$$A = \text{“}n\text{ is divisible by 6”},$$

and

$$B = \text{“}n\text{ is divisible by 3”}.$$

If (n) is divisible by (6), then (n) is divisible by (3). Therefore,

$$A \Rightarrow B.$$

So:

- being divisible by (6) is sufficient for being divisible by (3);
- being divisible by (3) is necessary for being divisible by (6).

However, being divisible by (3) is not sufficient for being divisible by (6). For example,

$$9$$

is divisible by (3), but it is not divisible by (6).
So we have:

$$A \Rightarrow B,$$

but not

$$B \Rightarrow A.$$

3.5 Logical Equivalence

Sometimes two statements imply each other.

Definition 6.

LogicalEquivalence

Statements (A) and (B) are logically equivalent *if*

$$A \Rightarrow B$$

and

$$B \Rightarrow A.$$

In this case we write

$$A \Leftrightarrow B.$$

We read this as:

"A if and only if B."

Logical equivalence means that (A) and (B) always have the same truth value. Either both are true or both are false.

Equivalently,

$$A \Leftrightarrow B$$

means that (A) is both necessary and sufficient for (B), and (B) is both necessary and sufficient for (A).

3.6 Truth Tables

Truth tables allow us to study logical statements by checking all possible truth values.

3.6.1 Implication

The truth table for $(A \Rightarrow B)$ is:

(A)	(B)	$(A \Rightarrow B)$
T	T	T
T	F	F
F	T	T
F	F	T

The only situation where $(A \Rightarrow B)$ is false is when (A) is true and (B) is false.

So the implication

$$A \Rightarrow B$$

is equivalent to saying:

not both A true and B false.

3.6.2 Equivalence

The truth table for $(A \Leftrightarrow B)$ is:

(A)	(B)	$(A \Leftrightarrow B)$
T	T	T
T	F	F
F	T	F
F	F	T

Equivalence is true exactly when (A) and (B) have the same truth value.

3.7 Converse, Inverse, and Contrapositive

Given an implication

$$A \Rightarrow B,$$

we can form three related statements.

Definition 7.

Converse

The converse *of* $(A \Rightarrow B)$ *is*

$$B \Rightarrow A.$$

Definition 8.

Inverse

The inverse *of* $(A \Rightarrow B)$ *is*

$$\neg A \Rightarrow \neg B.$$

Definition 9.

Contrapositive

The contrapositive *of* $(A \Rightarrow B)$ *is*

$$\neg B \Rightarrow \neg A.$$

The converse and inverse are not usually equivalent to the original implication. However, the contrapositive is always equivalent to the original implication.

Theorem 2.

Contrapositive

The implication

$$A \Rightarrow B$$

is logically equivalent to its contrapositive

$$\neg B \Rightarrow \neg A.$$

Proof. The implication $(A \Rightarrow B)$ is false only in one case: when (A) is true and (B) is false.

Now consider the contrapositive:

$$\neg B \Rightarrow \neg A.$$

This implication is false only when $(\neg B)$ is true and $(\neg A)$ is false. But $(\neg B)$ being true means that (B) is false, and $(\neg A)$ being false means that (A) is true.

So the contrapositive is false exactly when (A) is true and (B) is false.

Therefore, $(A \Rightarrow B)$ and $(\neg B \Rightarrow \neg A)$ are false in exactly the same situation. Hence they have the same truth value in every case, so they are logically equivalent. □

3.8 Examples

3.8.1 Geometric Example

Let

$$A = \text{“It is a square”},$$

and

$$B = \text{“It is a rectangle with equal sides”}.$$

Then:

- $(A \Rightarrow B)$: every square is a rectangle with equal sides;
- $(B \Rightarrow A)$: every rectangle with equal sides is a square.

Therefore,

$$A \Leftrightarrow B.$$

So being a square is equivalent to being a rectangle with equal sides.

3.8.2 Number Theory Example

Let

$$A = \text{“}n\text{ is divisible by 6”},$$

and

$$B = \text{“}n\text{ is divisible by 3”}.$$

Then:

$$A \Rightarrow B$$

is true, because every number divisible by (6) is also divisible by (3).
However,

$$B \Rightarrow A$$

is false, because (9) is divisible by (3), but not by (6).
So:

- (A) is sufficient for (B);
- (B) is necessary for (A);
- (A) and (B) are not logically equivalent.

3.8.3 Example with Inequalities

Let

$$A = \text{“}x > 2\text{”},$$

and

$$B = \text{“}x > 0\text{”}.$$

Assume (x) is a real number. If (x¿2), then certainly (x¿0). Therefore,

$$A \Rightarrow B.$$

So (x¿2) is sufficient for (x¿0), and (x¿0) is necessary for (x¿2).
But the converse is not true. For example,

$$x = 1$$

satisfies (x¿0), but does not satisfy (x¿2).
Thus,

$$A \Rightarrow B,$$

but not

$$B \Rightarrow A.$$

3.9 The Importance of the Domain

When working with logical statements in mathematics, we must know the domain of the variables.
For example, the statement

$$x^2 > 0$$

depends on what values of (x) are allowed.
If (x) is a nonzero real number, then

$$x^2 > 0$$

is true.
But if (x=0) is allowed, then the statement is false for (x=0).
Therefore, before deciding whether an implication is true or false, we must know the universe of objects we are discussing.

3.10 Exercises

1. Determine whether the following implication is true or false:
   “‘

   $$\text{“If } n \text{ is prime, then } n \text{ is odd.”}$$

   If it is false, give a counterexample.

2. Determine whether the following implication is true or false:

   $$\text{“If } n \text{ is prime and } n > 2\text{, then } n \text{ is odd.”}$$

   Identify the sufficient condition and the necessary condition.

3. Let

$$A = \text{“}x > 2\text{”}, \qquad B = \text{“}x > 0\text{”}.$$

   Write the converse, inverse, and contrapositive of $A \Rightarrow B$.

4. Give an example of a condition that is necessary but not sufficient.
5. Give two conditions that are both necessary for a third condition, but neither one alone is sufficient.
6. Construct the truth table for the statement

$$(A \Rightarrow B) \wedge (\neg A \Rightarrow B).$$

   What simpler statement is it logically equivalent to?

7. Let

$$A = \text{“}n \text{ is divisible by } 4\text{”}, \qquad B = \text{“}n \text{ is even”}.$$

   Decide whether $A \Rightarrow B$ is true. Then decide whether $B \Rightarrow A$ is true.

8. Give an example of two statements A and B such that

$$A \Leftrightarrow B.$$

   “‘

4 Algebraic Identities

4.1 Introduction

Algebraic identities are formulas that allow us to rewrite expressions in a more useful or factored form. They play a central role in number theory, especially when studying divisibility, prime numbers, and the structure of integers.

In this chapter we explore classical identities, their applications, and historical curiosities.

4.2 Difference of Powers

One of the most important identities is the factorization of a difference of powers:

Theorem 3 (Difference of Powers)**.** *For any integer x and positive integer n,*

$$x^n - 1 = (x-1)(x^{n-1} + x^{n-2} + \cdots + x + 1).$$

This identity is fundamental in the study of numbers of the form $x^n - 1$, especially when $x = 2$.

Example 4.1 (Factoring a Mersenne-Type Number)**.** *Consider the number*

$$2^{15} - 1.$$

Since $15 = 3 \cdot 5$*, we can apply the identity twice:*

$$2^{15} - 1 = (2^3 - 1)(2^{12} + 2^9 + 2^6 + 2^3 + 1).$$

Then factor $2^3 - 1$ *again:*

$$2^3 - 1 = (2-1)(2^2 + 2 + 1) = 7.$$

This method reveals the internal structure of the number without guessing prime factors.

4.3 Sum of Powers

Another classical identity is the sum of powers:

Theorem 4 (Sum of Powers)**.** *For any integer x and positive integer n,*

$$x^n + 1 = \begin{cases} (x+1)(x^{n-1} - x^{n-2} + \cdots - x + 1), & \textit{if } n \textit{ is odd} \\ \\ \textit{irreducible over the integers}, & \textit{if } n \textit{ is even.} \end{cases}$$

Example 4.2 (A Curious Identity)**.** *Since 5 is odd,*

$$2^5 + 1 = 33 = (2+1)(2^4 - 2^3 + 2^2 - 2 + 1).$$

This identity explains why some numbers of the form $2^n + 1$ factor, while others (the Fermat numbers) are suspected to be prime.

4.4 Historical Notes

Remark 2 (Euclid and Mersenne)**.** *Euclid already used the identity*

$$x^n - 1 = (x-1)(x^{n-1} + \cdots + 1)$$

in his proof that there are infinitely many primes. Much later, Marin Mersenne studied numbers of the form $2^p - 1$, now called Mersenne numbers*. Some of these are prime, and they produce the largest known primes in history.*

Remark 3 (Fermat's Curiosity)**.** *Pierre de Fermat conjectured that all numbers of the form*

$$F_n = 2^{2^n} + 1$$

were prime. He was correct for $n = 0, 1, 2, 3, 4$, but Euler disproved the conjecture by showing:

$$F_5 = 2^{32} + 1 = 4294967297 = 641 \cdot 6700417.$$

This is one of the most famous false conjectures in number theory.

4.5 Applications

- Factoring large integers using algebraic structure.
- Understanding prime candidates such as Mersenne and Fermat numbers.
- Simplifying expressions in modular arithmetic.
- Building conjectures about divisibility and primality.

4.6 Curiosities

- The largest known prime numbers are almost always of the form $2^p - 1$, where p is prime.
- The identity for $x^n - 1$ is used in cryptography, especially in algorithms involving cyclic groups.
- Some identities, like the sum of powers, behave differently depending on whether the exponent is odd or even — a subtle but important detail.

5 Limits, Infinitesimals, Continuity, and the Derivative

5.1 The Limit of a Variable

Definition 10. *A constant number a is said to be the* limit *of a variable x if, for every preassigned, arbitrarily small positive number ε, it is possible to indicate a value of the variable x such that all subsequent values of x satisfy the inequality*

$$|x - a| < \varepsilon.$$

Example

Consider the sequence $x_n = 1 + \frac{1}{n}$. We claim that $\lim_{n\to\infty} x_n = 1$.

Compute:

$$|x_n - 1| = \left|1 + \frac{1}{n} - 1\right| = \frac{1}{n}.$$

To make this less than ε, solve:

$$\frac{1}{n} < \varepsilon \quad \Rightarrow \quad n > \frac{1}{\varepsilon}.$$

Thus, for any $\varepsilon > 0$, choosing $n > \frac{1}{\varepsilon}$ guarantees

$$|x_n - 1| < \varepsilon.$$

Hence the sequence converges to 1.

5.2 The Limit of a Function

Definition 11. *Let $f(x)$ be a function. We say that the limit of $f(x)$ as x approaches a is b, written*

$$\lim_{x\to a} f(x) = b,$$

if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that whenever, for all x, $0 < |x - a| < \delta$, the inequality

$$|f(x) - b| < \varepsilon$$

holds.

This definition formalizes the idea that $f(x)$ can be made arbitrarily close to b by taking x sufficiently close to a.

5.3 Infinitesimals

Definition 12. *A function $\alpha = \alpha(x)$ is called* infinitesimal *as $x \to a$ (or as $x \to \infty$) if*

$$\lim_{x\to a} \alpha(x) = 0 \quad \textit{or} \quad \lim_{x\to\infty} \alpha(x) = 0.$$

Theorem 1

If $y = b + \alpha(x)$, where α is infinitesimal, then

$$\lim y = b.$$

Conversely, if $\lim y = b$, then y may be written as $y = b + \alpha$, where α is infinitesimal.

Proof. From $y = b + \alpha$,

$$|y - b| = |\alpha|.$$

Given any $\varepsilon > 0$, since α is infinitesimal, $|\alpha| < \varepsilon$, hence $|y - b| < \varepsilon$.

Theorem 2

If $\alpha(x)$ approaches 0 as $x \to a$ (or $x \to \infty$) and never becomes zero, then

$$y = \frac{1}{\alpha(x)} \to \infty.$$

Proof. Given any large $M > 0$, choose $|\alpha| < \frac{1}{M}$. Then

$$\frac{1}{|\alpha|} > M.$$

Theorem 3

The algebraic sum of a finite number of infinitesimals is an infinitesimal.

Proof. Let $u(x) = \alpha(x) + \beta(x)$, where both α and β are infinitesimal. Then for sufficiently small x,

$$|\alpha| < \frac{\varepsilon}{2}, \qquad |\beta| < \frac{\varepsilon}{2}.$$

Thus,

$$|u| = |\alpha + \beta| \leq |\alpha| + |\beta| < \varepsilon.$$

Theorem 4

The product of an infinitesimal $\alpha(x)$ and a bounded function $z(x)$ is infinitesimal.

Proof. If $|z(x)| < M$ and $|\alpha| < \frac{\varepsilon}{M}$, then

$$|z(x)\alpha(x)| < M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

5.4 Continuity of Functions

Definition 13. *A function $y = f(x)$ is* continuous *at x_0 if it is defined in a neighborhood of x_0 and*

$$\lim_{x \to x_0} f(x) = f(x_0).$$

Equivalently,

$$\lim_{\Delta x \to 0} [f(x_0 + \Delta x) - f(x_0)] = 0.$$

Since

$$x_0 = \lim_{x \to x_0} x,$$

we may write

$$\lim_{x \to x_0} f(x) = f(\lim_{x \to x_0} x).$$

Thus, for continuous functions, limits may be computed by direct substitution.

Theorem 1

If $f_1(x)$ and $f_2(x)$ are continuous at x_0, then $f_1(x) + f_2(x)$ is also continuous at x_0.

5.5 Continuity on an Interval

Definition 14. *A function $f(x)$ is continuous on an interval (a, b) if it is continuous at every point of the interval.*

5.6 Properties of Continuous Functions

Theorem 1 (Extreme Value Theorem)

If $f(x)$ is continuous on a closed interval $[a, b]$, then there exist points x_1 and x_2 in $[a, b]$ such that

$$f(x_1) \geq f(x) \quad \text{and} \quad f(x_2) \leq f(x)$$

for all x in $[a, b]$. That is, f attains its maximum and minimum on the interval.

5.7 Definition of the Derivative

Let $y = f(x)$ be continuous on an interval. For an increment Δx, define

$$\Delta y = f(x + \Delta x) - f(x).$$

Form the ratio

$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

Definition 15 (Derivative)**.** *If the limit*

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

exists, it is called the derivative *of $y = f(x)$ and is denoted by*

$$y' = f'(x).$$